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On fuzzy submodules and fuzzy Nakayama's lemma

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ABSTRACT

Let R be a commutative ring with identity, ξ a fuzzy ideal of R, and let M be an R-module. Let λ be a finitely generated fuzzy submodule of M such that $\xi \odot \lambda = \lambda$. It is proved that there exist $a \in \xi_1$ such that $1_{\mathbb{N}(1+a)} \odot \lambda = 1_\theta$. Then, using this result, a proof of fuzzy version of Nakayama's lemma is given.

1. Introduction

In recent decades, the framework of fuzzy set theory, introduced by L. Zadeh [1], has become a central tool in extending classical mathematical structures to settings involving uncertainty or gradual membership. Within algebra, this has led to the development of fuzzy structures where the crisp inclusion relations of classical algebra are replaced by membership functions that take values in the unit interval I or more generally in the complete Heyting algebra $(L, \vee, \wedge, \leq, 0, 1)$. Motivated by this trend, several classical results in algebra have been revisited within the fuzzy framework [2–4].

Fuzzy analogues of concepts such as primary decomposition, Jacobson and nil radicals, finitely generated submodules, and fractions formation have been established, and various extensions of classical theorems have been explored (see for example [5–10]). In this direction, using a structural result of Pan [11, Theorem 2.1] for a fuzzy submodule of a finitely generated module M, a special fuzzy version of Nakayama's lemma is formulated and studied in [12] There, it is assumed that M itself is finitely generated and consequently every fuzzy submodule μ of M is generated by a set of generators of M, in which μ has the same value $t \in [0, 1]$ in all of the generators (see also [13], and [14] for a proof of Nakayama's lemma with an extra assumption on the set of generators of the fuzzy submodule). Afterwards, a proof of fuzzy Nakayama's lemma is given in [15] using the support of fuzzy product of a fuzzy ideal and a fuzzy submodule.

The aim of this paper is a study of finitely generated fuzzy submodules (and finitely generated fuzzy quotient modules) in which M itself is not necessarily assumed to be finitely generated, but the fuzzy submodule μ is. Then, using the results presented, we conclude the fuzzy version of the Nakayama's lemma. To be precise, assume that R is a commutative ring with identity and that M is an R-module with zero element θ. Let λ be a finitely generated fuzzy submodule of M and let ξ be a fuzzy ideal of R such that $\xi \odot \lambda = 1_\theta$. It is proved that there exists $a \in \xi_1$ such that $1_{\mathbb{N}(1+a)} \odot \lambda = 1_\theta$. Using this fact the fuzzy version of Nakayama's lemma is obtained (Theorem 3.2). To obtain these, we need some elementary, but useful results which we bring them all in Section 2.

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2. Basic notations and auxiliary results

In this section we introduce some notations and prove useful lemmas we need in the sequel. For any unexplained definition we refer to the textbook [16]. Let R be a commutative ring with identity $1 \neq 0$, and let M be a unital R-module with zero element θ. Let $\mu : M \to [0,1] =: I$ be a fuzzy submodule of M, so that $\mu(\theta) = 1$, and for each $r \in R, x, y \in M$ one has $\mu(rx) \geq \mu(x)$ and $\mu(x+y) \geq \mu(x) \wedge \mu(y)$. It is known that, for each $t \in I$, $\mu_t := \{x \in M \mid \mu(x) \geq t\}$ (the t-level set of μ) and $\mu^* = \cup_{t>0}\mu_t$ (the support of μ) are (crisp) submodules of M [16]. We assume that fuzzy ideals we consider are normalized, that is, for a typical fuzzy ideal $\xi : R \to I$ of R, $\xi(0) = 1$. As usual we will use the symbols $\vee$ and $\wedge$ to denote the supremum and infimum in I, respectively. For two fuzzy subsets η, μ of M, $\eta \subseteq \mu$, if $\eta(x) \leq \mu(x)$ for each $x \in M$, and $\eta \subset \mu$ if $\eta \subseteq \mu$ and there exists $x \in M$ such that $\eta(x) < \mu(x)$. Let Y be a subset of M and let $t \in I \setminus \{0\}$. The map

$$t_Y(x) = \begin{cases} t & \text{if} \quad x \in Y \\ 0 & \quad \text{otherwise,} \end{cases}$$

is a fuzzy subset of M. In particular, if $y \in M, t_y := t_{\{y\}}$ is called a fuzzy singleton (point) in M with value t, and for a fuzzy subset λ of M we will write $t_y \in \lambda \Leftrightarrow \lambda(y) \geq t$. Recall that a fuzzy submodule λ of M is called (fuzzy) zero, if $\lambda = 1_\theta$, and it is fuzzy principle if $\lambda = \langle t_x \rangle = 1_\theta \cup t_{Rx}$ for some $t \in I$ and some $x \in M$, where Rx is the cyclic submodule of M generated by x (see [16, p. 140]). Similar mentioned notations are assumed for fuzzy ideals, and, in particular, a fuzzy ideal ξ of R is principal if it is of the form $\langle t_r \rangle = 1_0 \cup t_{Rr}$ for some $t \in I$ and some $r \in R$.

For a fuzzy subset ς of R and for a fuzzy subset α of M, $\varsigma \odot \alpha$ is defined as

$$(\varsigma \odot \alpha)(x) = \vee\{\wedge_{i=1}^k (\varsigma(r_i) \wedge \alpha(x_i)) | k \in \mathbb{N}, r_i \in R, x_i \in M, x = \Sigma_{i=1}^k r_i x_i\},$$

which is a fuzzy subset of M. In particular, for $t, s \in I, a, b \in R$, $y \in M$

$$(t_a \odot \alpha)(x) = \vee\{\wedge_{i=1}^k (t \wedge \alpha(x_i)) | k \in \mathbb{N}, x_i \in M, x = a\Sigma_{i=1}^k x_i\},$$

$$(t_a \odot s_y)(x) = (t \wedge s)_{\mathbb{N}ay}(x) = \begin{cases} t \wedge s & \text{if } x = kay, \text{ for some } k \in \mathbb{N}, \\ 0 & \quad \text{otherwise.} \end{cases}$$

for each $x \in M$. In particular

$$(t_a \odot s_b) = (t \wedge s)_{\mathbb{N}ab}.$$

We note that, when ξ is a fuzzy ideal of R and λ is a fuzzy submodule of M, then $\xi \odot \lambda$ is a fuzzy submodule of M contained in λ.

A fuzzy submodule λ of M is said to be finitely generated if there exists a finite set of fuzzy singletons $(t_1)_{y_1}, \dots, (t_n)_{y_n}$ such that $\lambda = \langle (t_1)_{y_1}, \dots, (t_n)_{y_n} \rangle$, where $t_i \in I$ and $y_i \in M$, $1 \leq i \leq n$. This equivalently means that there exists a fuzzy subset α of M with finite support $\alpha^* = \{y_1, \dots, y_n\}$ and with $\mathrm{Im}(\alpha) = \{t_1, \dots, t_n\}$ such that $\lambda = \langle \alpha \rangle$. Note that in this case $\lambda = 1_\theta \cup (1_R \odot \alpha)$. Thus, for each $\theta \neq x \in M$,

$$\begin{aligned} \lambda(x) &= \vee\{\wedge_{i=1}^k \alpha(x_i) | k \in \mathbb{N}, 1 \leq i \leq k, r_i \in R, x_i \in M, x = \Sigma_{i=1}^k r_i x_i\} \\ &= \vee\{\wedge_{j=1}^\ell t_{i_j} | 1 \leq \ell \leq n, 1 \leq i_1 < i_2 < \cdots < i_\ell \leq n, r_{i_j} \in R, x = \Sigma_{j=1}^\ell r_{i_j} y_{i_j}\}, \ (\dagger) \end{aligned}$$

and $\lambda(\theta) = 1$. As a convention, the fuzzy submodule generated by the null set is 1_θ.

Now suppose that η and μ are fuzzy submodules of M. Then as usual we can consider the quotient fuzzy submodule $\frac{\mu}{\eta}$ as the fuzzy submodule $\frac{\mu+\eta}{\eta}$ of the R-module $\hat{M} = M/\eta^*$, which is defined by

$$(\frac{\mu}{\eta})(\hat{x}) = \vee\{(\mu+\eta)(x+e) | e \in \eta^*\}, \qquad (*)$$

for each $\hat{x} = x + \eta^* \in M$, where $\mu + \eta$ is the usual fuzzy sum of μ and η. The particular case is when $\eta \subseteq \mu$, so that

$$(\frac{\mu}{\eta})(\hat{x}) = \vee\{\mu(x+e) | e \in \eta^*\}. \qquad (\ddagger)$$

With this notations we have the following lemma.

Lemma 2.1. *Let μ be a finitely generated fuzzy submodule of M and let η be a submodule of M such that $\eta \subseteq \mu$. Then, $\frac{\mu}{\eta}$ is a finitely generated fuzzy submodule of M/η^*.*

Proof. Assume that $\mu = \langle (t_1)_{y_1}, \dots, (t_n)_{y_n} \rangle$. We show that $\frac{\mu}{\eta} = \langle (t_1)_{\hat{y_1}}, \dots, (t_n)_{\hat{y_n}} \rangle$. So, let $x \in M$. Using ($\ddagger$), $\dagger$, and combining the joins we have

$$\begin{aligned} \frac{\mu}{\eta}(\hat{x}) &= \vee_{e \in \eta^*} \mu(x+e) \\ &= \vee\{\wedge_{j=1}^\ell t_{i_j} | 1 \leq i_1 < i_2 < \cdots < i_\ell \leq n, r_{i_j} \in R, e \in \eta^*, x+e = \Sigma_{j=1}^\ell r_{i_j} y_{i_j}\} \\ &= \vee\{\wedge_{j=1}^\ell t_{i_j} | 1 \leq i_1 < i_2 < \cdots < i_\ell \leq n, r_{i_j} \in R, \hat{x} = \Sigma_{j=1}^\ell r_{i_j} \hat{y_{i_j}}\}. \end{aligned}$$

This, together with the fact that $\frac{\mu}{\eta}(\eta^*) = 1$, means that

$$\frac{\mu}{\eta} = \langle (t_1)_{\hat{y}_1}, (t_2)_{\hat{y}_2}, \dots, (t_n)_{\hat{y}_n} \rangle.$$

Hence $\frac{\mu}{\eta}$ is a finitely generated fuzzy submodule of $\hat{M}$, as required. □

We need one more observation on finitely generated fuzzy submodules. Suppose that $n \geq 1$ and that the fuzzy submodule μ is generated by $n+1$ fuzzy singletons $(a_1)_{y_1}, \dots, (a_n)_{y_n}, (a_{n+1})_{y_{n+1}}$, and let $a_{n+1} = \vee\{a_i \mid 1 \leq i \leq n+1\}$. We claim that the fuzzy submodule $\frac{\mu}{\langle (a_{n+1})_{y_{n+1}} \rangle}$ of M/Ry_{n+1} is generated by the n fuzzy singletons $(a_1)_{y_1+Ry_{n+1}}, \dots, (a_n)_{y_n+Ry_{n+1}}$. To see this, for simplicity, we set $\nu = \langle (a_{n+1})_{y_{n+1}} \rangle$ so that $\nu^* = Ry_{n+1}$, $\bar{x} = x + \nu^*$ for each $x \in M$. Let $\eta = \langle (a_1)_{\bar{y_1}}, \dots, (a_1)_{\bar{y_n}} \rangle$. Note that for each $x \in M$, there exists $1 \leq \ell \leq n+1$ with $1 \leq i_1 < \dots < i_\ell \leq n+1$ and there exists $r_{i_j} \in R$ with $1 \leq j \leq \ell$ such that $\Sigma_{i=1}^{\ell} r_{i_j} y_{i_j} = x$ if and only if there exists $1 \leq \ell \leq n$ with $1 \leq i_1 < \dots < i_\ell \leq n$ and there exists $s_{i_j} \in R$ with $1 \leq j \leq \ell$ such that $\Sigma_{i=1}^{\ell} s_{i_j} \bar{y_{i_j}} = \bar{x}$. Thus, with the choice of a_{n+1}, we have

$$\begin{aligned}
\mu(x) &= \vee\{\wedge_{j=1}^{\ell} a_{i_j} \mid 1 \leq \ell \leq n+1, 1 \leq i_1 < \dots < i_\ell \leq n+1, r_{i_j} \in R, \Sigma_{j=1}^{\ell} r_{i_j} y_{i_j} = x\} \\
&= \vee\{\wedge_{j=1}^{\ell} a_{i_j} \mid 1 \leq \ell \leq n, 1 \leq i_1 < i_2 < \dots < i_\ell \leq n, r_{i_j} \in R, \Sigma_{j=1}^{\ell} r_{i_j} \bar{y_{i_j}} = \bar{x}\} \\
&= \eta(\bar{x}). \qquad (\natural)
\end{aligned}$$

Hence

$$\begin{aligned}
(\mu/\nu)(\bar{x}) &= \vee\{\mu(x + rx_{n+1}) \mid r \in R\} \\
&= \vee\{\eta(x + ry_{n+1} + \nu^*) \mid r \in R\} \\
&= \vee\{\eta(\bar{x}) \mid r \in R\} = \eta(\bar{x}).
\end{aligned}$$

With this observations we obtain the following lemma.

Lemma 2.2. *Let $n \geq 1$ and let μ be a finitely generated fuzzy submodule of M which is generated by the fuzzy singletons $(a_1)_{y_1}, \dots, (a_{n+1})_{y_{n+1}}$. Then, there exists some $1 \leq j \leq n+1$ such that the fuzzy submodule $\eta = \frac{\mu}{\langle (a_j)_{y_j} \rangle}$ of M/Ry_j can be generated by n fuzzy singletons.*

For a fuzzy ideal ξ of R and fuzzy submodules μ and η of M, one can define $\xi \odot \frac{\mu}{\eta}$, a fuzzy submodule of M/η^*. The following lemma shows that this is somehow consistence with the corresponding definition in crisp module case.

Lemma 2.3. *Let ξ be a fuzzy ideal of R and let μ and η be fuzzy submodules of M. Then, $\xi \odot \frac{\mu}{\eta} = \frac{\xi \odot \mu}{\eta}$.*

Proof. By $(*)$ we need to show that for each $x \in M$,

$$(\xi \odot \frac{\mu}{\eta})(x + \eta^*) = \frac{\xi \odot \mu + \eta}{\eta}(x + \eta^*).$$

To do this, again set $\hat{x} = x + \eta^* \in \hat{M} = M/\eta^*$. We have

$$\begin{aligned}
&(\xi \odot \frac{\mu}{\eta})(\hat{x}) = \vee\{\wedge_{i=1}^{k} (\xi(r_i) \wedge \frac{\mu}{\eta}(\hat{x_i})) \mid k \in \mathbb{N}, r_i \in R, x_i \in M, \Sigma_{i=1}^{k} r_i \hat{x_i} = \hat{x}\} \\
&= \vee\{\wedge_{i=1}^{k} (\xi(r_i) \wedge \vee \mu(x_i + e_i)) \mid k \in \mathbb{N}, r_i \in R, x_i \in M, e_i \in \eta^*, \Sigma_{i=1}^{k} r_i \hat{x_i} = \hat{x}\} \\
&= \vee\{\wedge_{i=1}^{k} (\xi(r_i) \wedge \mu(x_i + e_i)) \mid k \in \mathbb{N}, r_i \in R, , x_i \in M, e_i \in \eta^*, \Sigma_{i=1}^{k} r_i x_i - x \in \eta^*\} \\
&= \vee\{\wedge_{i=1}^{k} (\xi(r_i) \wedge \mu(x_i + e_i)) \mid k \in \mathbb{N}, r_i \in R, x_i \in M, e_i \in \eta^*, \Sigma_{i=1}^{k} r_i (x_i + e_i) - x \in \eta^*\} \\
&= \vee\{\wedge_{i=1}^{k} (\xi(r_i) \wedge \mu(z_i)) \mid k \in \mathbb{N}, r_i \in R, , x_i \in M, \Sigma_{i=1}^{k} r_i z_i - x \in \eta^*\} \qquad (\flat),
\end{aligned}$$

where the third equality holds because finite meets distribute over arbitrary joins, and for the last equality note that $\Sigma r_i(x_i + e_i) - x \in \eta^*$ if and only if $\Sigma r_i x_i - x \in \eta^*$. On the other hand,

$$\begin{aligned}
&\frac{\xi \odot \mu + \eta}{\eta}(\hat{x}) = \vee\{(\xi \odot \mu + \eta)(x + e) \mid e \in \eta^*\} \\
&= \vee\{(\xi \odot \mu)(y) \wedge \eta(z) \mid e \in \eta^*, y, z \in M, x + e = y + z\} \\
&= \vee\{\wedge_{i=1}^{k} \xi(r_i) \wedge \mu(z_i) \wedge \eta(z) \mid k \in \mathbb{N}, r_i \in R, e \in \eta^*, y, z, z_i \in M, x + e = y + z, \Sigma_{i=1}^{k} r_i z_i = y\} \\
&= \vee\{\wedge_{i=1}^{k} \xi(r_i) \wedge \mu(z_i) \wedge \eta(z) \mid k \in \mathbb{N}, r_i \in R, e, z \in \eta^*, y, z_i \in M, x + e = y + z, \Sigma_{i=1}^{k} r_i z_i = y\} \\
&= \vee\{\wedge_{i=1}^{k} \xi(r_i) \wedge \mu(z_i) \mid k \in \mathbb{N}, r_i \in R, y, z_i \in M, \Sigma r_i z_i - x \in \eta^*\}, \qquad (\sharp)
\end{aligned}$$

where the forth and fifth equalities is because z appears inside $\eta(-)$, the condition "$\eta(z)$ large" is only relevant when $z \in \eta_1 \subseteq \eta^*$; and then effectively, the tuples counted are those with $\Sigma r_i z_i - x \in \eta^*$. Thus, from $(\flat)$ and $(\sharp)$ we obtain the desired equality. □

Lemma 2.4. *Let ξ be a fuzzy ideal of R and let μ, η be fuzzy submodules of M. Then, $\xi \odot \frac{\mu}{\eta} \subseteq 1_\theta$ if and only if $\xi \odot \mu \subseteq 1_{\eta^*}$.*

Proof. First, assume that $\xi \odot \frac{\mu}{\eta} \subseteq 1_\theta$. This means that for each $\hat{x} \in \hat{M} = M/\eta^*$ we have $(\xi \odot \frac{\mu}{\eta})(\hat{x}) \leq 1_\theta(\hat{x})$, and $(\xi \odot \frac{\mu}{\eta})(\hat{x}) = 0$ whenever $x \notin \eta^*$. Therefore, for each $x \notin \eta^*$

$$\begin{aligned}
&(\xi \odot \frac{\mu}{\eta})(\hat{x}) = \vee\{\wedge_{i=1}^{k} (\xi(r_i) \wedge (\frac{\mu}{\eta})(\hat{x_i})) \mid k \in \mathbb{N}, r_i \in R, x_i \in M, \Sigma_{i=1}^{k} r_i x_i - x \in \eta^*\} \\
&= \vee\{\wedge_{i=1}^{k} (\xi(r_i) \wedge \vee_{e_i \in \eta^*} \mu(x_i + e_i)) \mid k \in \mathbb{N}, r_i \in R, x_i \in M, e_i \in \eta^*, \Sigma_{i=1}^{k} r_i x_i - x \in \eta^*\}
\end{aligned}$$

$$= \vee\{\wedge_{i=1}^{k}(\eta(r_i) \wedge \mu(x_i + e_i))|k \in \mathbb{N}, r_i \in R, x_i \in M, e_i \in \eta^*, \Sigma_{i=1}^{k} r_i x_i - x \in \eta^*\}$$
$$= \vee\{\wedge_{i=1}^{k}(\xi(r_i) \wedge \mu(x_i + e_i))|k \in \mathbb{N}, r_i \in R, x_i \in M, e_i \in \eta^*, \Sigma_{i=1}^{k} r_i (x_i + e_i) - x \in \eta^*\}$$
$$= \vee\{\wedge_{i=1}^{k}(\xi(r_i) \wedge \mu(z_i)|k \in \mathbb{N}, r_i \in R, z_i \in M, \Sigma_{i=1}^{k} r_i z_i - x \in \eta^*\}.$$

Since $(\xi \odot \frac{\mu}{\eta})(\hat{x}) = 0$ whenever $x \notin \eta^*$, thus each inner meet must be 0. Hence all nonzero contributions occur only when $x \in \eta^*$. Thus,

$$(\xi \odot \mu)(x) = \vee\{\wedge_{i=1}^{k}(\xi(r_i) \wedge \mu(x_i))|k \in \mathbb{N}, r_i \in R, x_i \in M, \Sigma_{i=1}^{k} r_i x_i = x\} \leq 1_{\eta^*}(x),$$

and we conclude that $\xi \odot \mu \subseteq 1_{\eta^*}$.

Conversely, assume that $\xi \odot \mu \subseteq 1_{\eta^*}$. This means that for each $x \notin \eta^*$,

$$\vee\{\wedge_{i=1}^{k}(\xi(r_i) \wedge \mu(x_i))|k \in \mathbb{N}, r_i \in R, x_i \in M, \Sigma_{i=1}^{k} r_i x_i = x\} = (\xi \odot \mu)(x) = 0,$$

and thus for each $k \in \mathbb{N}, r_i \in R, x_i \in M$ with $\Sigma_{i=1}^{k} r_i x_i = x$, we have

$$\wedge_{i=1}^{k}(\xi(r_i) \wedge \mu(x_i)) = 0. \qquad (\flat\flat)$$

Now,

$$(\xi \odot \frac{\mu}{\eta})(\hat{x}) = \vee\{\wedge_{i=1}^{k}(\xi(r_i) \wedge \frac{\mu}{\eta}(\hat{x}_i))|k \in \mathbb{N}, r_i \in R, x_i \in M, \Sigma_{i=1}^{k} r_i \hat{x}_i = \hat{x} \in \eta^*\}$$
$$= \vee\{\wedge_{i=1}^{k}(\xi(r_i) \wedge \mu(x_i + e_i))|k \in \mathbb{N}, r_i \in R, x_i \in M, \Sigma_{i=1}^{k} r_i x_i - x \in \eta^*,\ e_i \in \eta^*\}$$
$$= \vee\{\wedge_{i=1}^{k}(\xi(r_i) \wedge \mu(z_i))|k \in \mathbb{N}, r_i \in R, z_i \in M, \Sigma_{i=1}^{k} r_i z_i - x \in \eta^*\}.$$
$$= \vee\{\wedge_{i=1}^{k}(\xi(r_i) \wedge \mu(z_i))|k \in \mathbb{N}, r_i \in R, z_i \in M, \Sigma_{i=1}^{n} r_i z_i = x + u, u \in \eta^*\}.$$

If $x \notin \eta^*$, then each sum $\sum_{i=1}^{n} r_i z_i$ appearing in the last expression is not in η^*. Hence, by $(\flat\flat)$, each meet term is 0. Hence $(\xi \odot \frac{\mu}{\eta})(\hat{x}) = 0$ for all $x \notin \eta^*$, which means that

$$\xi \odot \frac{\mu}{\eta} \subseteq 1_\theta.$$

The proof now is complete. □

3. A proof of fuzzy Nakayama's lemma

The aim of this section is to prove the fuzzy Nakayama's lemma. We begin with the following key theorem.

Theorem 3.1. *Let λ be a finitely generated fuzzy submodule of M and let ξ be an ideal of R such that $\xi \odot \lambda = \lambda$. Then there exists $a \in \xi_1$ such that $1_{\mathbb{N}(1+a)} \odot \lambda = 1_\theta$.*

Proof. We use induction on the number n of singletons required to generated λ. If $n = 0$, then $\lambda = 1_\theta$ and there is nothing to prove. So, assume that the result has been proved for all fuzzy submodules which can be generated by $n \geq 1$ singletons. Assume also that μ is generated by $n+1$ points $(t_1)_{y_1}, \ldots, (t_n)_{y_n}, (t)_y$ and that $\xi \odot \mu = \mu$. We may assume that $t \geq t_i$ for $1 \leq i \leq n$. Let $\eta = \frac{\mu}{\nu}$, where $\nu = \langle (t)_y \rangle$. Thus, according to Lemma 2.2 and the notations before it, η can be generated by the n points $(t_1)_{\bar{y}_1}, \ldots, (t_1)_{\bar{y}_n}$. Furthermore, by Lemma 2.3 and definition $(*)$, we see that $\xi \odot \frac{\mu}{\nu} = \frac{\xi \odot \mu + \nu}{\nu} = \frac{\mu + \nu}{\nu} = \frac{\mu}{\nu}$. Thus, by the inductive hypothesis, there exists $b \in \xi_1$ such that $1_{\mathbb{N}(1+b)} \odot \frac{\mu}{\nu} = 1_\theta$. Thus, by Lemma 2.4, we have $1_{\mathbb{N}(1+b)} \odot \mu \subseteq 1_{\nu^*} (= 1_{Ry})$. Hence,

$$1_{\mathbb{N}(1+b)} \odot \mu = 1_{\mathbb{N}(1+b)} \odot \xi \odot \mu = \xi \odot 1_{\mathbb{N}(1+b)} \odot \mu \subseteq \xi \odot 1_{\nu^*}.$$

This, in particular, gives that $(1_{\mathbb{N}(1+b)} \odot \mu)((1+b)y) \leq (\xi \odot 1_{\nu^*})((1+b)y)$. This, in turn, means and implies that

$$\mu(y) \leq \vee\{\wedge_{i=1}^{k}\mu(x_i)|k \in \mathbb{N}, x_i \in M, \Sigma_{i=1}^{k}(1+b)x_i = (1+b)y\}$$
$$\leq \vee\{\wedge_{i=1}^{k}\xi(r_i)|k \in \mathbb{N}, r_i \in R, \Sigma_{i=1}^{k} r_i y = (1+b)y\}.$$

Thus, for each ϵ, for example $\epsilon = 1 - \mu(y)$, there exists a decomposition $\Sigma_{i=1}^{k} r_i y = (1+b)y$ such that $1 = \mu(y) - \epsilon \leq \wedge_{i=1}^{k}\xi(r_i)$. Hence $r_j \in \xi_1$ for all $j = 1, \ldots, k$. Then we can define $c = -\Sigma_{i=1}^{k} r_i \in \xi_1$ and with this c we have $1_{\mathbb{N}(1+b+c)} \odot 1_{\nu^*} = 1_\theta$. Thus we have $1_{\mathbb{N}(1+b+c)} \odot 1_{\mathbb{N}(1+b)} \odot \mu \subseteq 1_{\mathbb{N}(1+b+c)} \odot \xi \odot 1_{\nu^*} \subseteq 1_{\mathbb{N}(1+b+c)} \odot 1_{\nu^*} = 1_\theta$, where $1_{\mathbb{N}(1+b+c)} \odot 1_{\mathbb{N}(1+b)} = 1_{\mathbb{N}(1+a)}$ with $a \in \xi_1$. This completes the inductive step. □

Following [6], a fuzzy ideal ξ of R is called a maximal fuzzy ideal maximal if it is maximal in the lattice of all nonconstant fuzzy ideals of R. Equivalently, ξ is a maximal fuzzy ideal if and only if $\xi(1) < 1$ and whenever $\xi(b) < 1$ for some $b \in R$, then $\xi(1 + rb) = 1$ for some $r \in R$ [6]. It has been seen in [17] that a fuzzy ideal ξ is maximal fuzzy ideal if and only if $\xi_1 \subseteq R$ is a maximal ideal. It follows that maximal fuzzy ideals are exactly characteristic functions of maximal ideals. The fuzzy Jacobson radical of R, we denote it by $FJ(R)$, is the intersection of all fuzzy maximal ideals of R. Thus for each $r \in R$, $FJ(R) = \wedge\{\xi|\xi \text{ is a maximal fuzzy ideal}\} = \wedge\{1_{\mathfrak{m}}|\mathfrak{m} \text{ is a maximal ideal}\} = 1_{J(R)}$, where $J(R)$ is the Jacobson radical of R. It is known that if a nonconstant fuzzy ideal ξ of R is contained in $FJ(R)$, then for each $a \in \xi_1$, the element $1 + a$ is invertible in R [6, 2.6. Theorem]. It is also known that $a \in R$ is invertible if and only if the element $1_a + FJ(R)$ is invertible in the fuzzy quotient ring $R/FJ(R)$ (see [6, 2.11. Theorem] and see also [16, pages. 83,84] for the quotient ring of R by a fuzzy ideal ζ).
Now, we can state and prove the fuzzy version of the Nakayama's lemma.

Theorem 3.2. *Let μ be a finitely generated fuzzy submodule of the R-module M and let ξ be a fuzzy ideal of R such that $\xi \subseteq FJ(R)$. Assume that $\xi \odot \mu = \mu$. Then, $\mu = 1_\theta$.*

Proof. By Theorem 3.1 there exists $a \in \xi_1 \subseteq FJ(R)_1$ such that $1_{\mathbb{N}(1+a)} \odot \mu = 1_\theta$. By the observations in the previous paragraph we know that $1 + a$ is invertible in R. So let $u \in R$ such that $(1 + a)u = 1$. Then,

$$\mu = 1_1 \odot \mu \subseteq 1_{\mathbb{N}.1} \odot \mu = 1_u \odot 1_{1+a} \odot \mu \subseteq 1_u \odot 1_\theta = 1_\theta,$$

as desired. □

Corollary 3.3. *Let μ be a finitely generated fuzzy submodule of M and let η be a submodule of M such that $\eta \subseteq \mu$. Let ξ be a fuzzy ideal of R contained in $FJ(R)$. Assume that $\xi \odot \mu + \eta = \mu$. Then, $\mu = 1_{\eta^*}$.*

Proof. Assume that μ is generated by the fuzzy singletons $(a_1)_{y_1}, \dots, (a_n)_{y_n}$. Then, by Lemma 2.1, $\frac{\mu}{\eta}$ is a finitely generated fuzzy submodule of $\hat{M} = M/\eta^*$ that can be generated by the fuzzy singletons $(a_1)_{\hat{y}_1}, \dots, (a_n)_{\hat{y}_n}$, where $\hat{y}_i = y_i + \eta^*$. Furthermore, by Lemma 2.3, we have $\xi \odot \frac{\mu}{\eta} = \frac{\xi \odot \mu + \eta}{\eta} = \frac{\mu}{\eta}$. Thus, applying Theorem 3.2 to $\frac{\mu}{\eta}$, we see that $\frac{\mu}{\eta} = 1_\theta$. Thus, by Lemma 2.4, $\mu = 1_{\eta^*}$ as desired. □

4. Conclusion and possible future perspectives

In this paper, we studied some properties of finitely generated fuzzy submodules. Among other results, it was seen that for a finitely generated fuzzy submodule λ of an R-module M, and for a fuzzy ideal ξ of R satisfying $\xi \odot \lambda = \lambda$, there exists $a \in \xi_1$ such that $1_{\mathbb{N}(1+a)} \odot \lambda = 1_\theta$. This resulted in a new proof of fuzzy Nakayama's Lemma. Using the results appeared and the previously known results on fuzzy fractions formation, one can focus in future on the support of fuzzy quotient modules like $\frac{\lambda}{\xi \odot \lambda}$, where λ is a finitely generated fuzzy submodule of M and ξ is an ideal of R.

CRediT authorship contribution statement

Naser Zamani: Writing – review & editing, Conceptualization.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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